DATA ACQUISITION

PHX-DAQ 24

4-channel 24-bit sensor and temperature datalogger

Ethernet-connected logger for strain gauges, load cells, pressure transducers and other bridge-type sensors. Works with PHX-Sense Studio, PLCs, SCADA and NI LabVIEW.



Pre-production unit shown.

KEY FEATURES

- **4 independent 24-bit inputs**, one converter per channel, for bridge-type sensors
- **Programmable gain 1× to 128×**, set per channel
- **4 temperature inputs** for NTC probes, for on-site temperature compensation
- **1 to 100 Hz** selectable output rate
- **Ready-to-use mV/V values** for PLC and SCADA, averaged to remove mains hum NEW
- **Sensor check**: detects a disconnected sensor or broken wire on each channel NEW
- **Up to 3 PCs** receive full-rate data at once NEW
- **Ethernet 10/100** with Modbus TCP and a continuous data stream
- **Automatic discovery** on the network; fixed factory IP, DHCP or static
- **Built-in Wi-Fi access point** for on-site setup
- **Authenticated** configuration and firmware updates
- **Network firmware updates** with automatic rollback
- **Self-reporting health**: uptime and restart counters
- **Reset button** restores factory defaults NEW
- **24 V DC** supply with input protection

COMPATIBLE SENSORS

Strain gauges (¼, ½, full bridge) · Load cells · Pressure transducers · Force & torque sensors · Other mV-level bridge sensors · NTC temperature probes

WORKS WITH

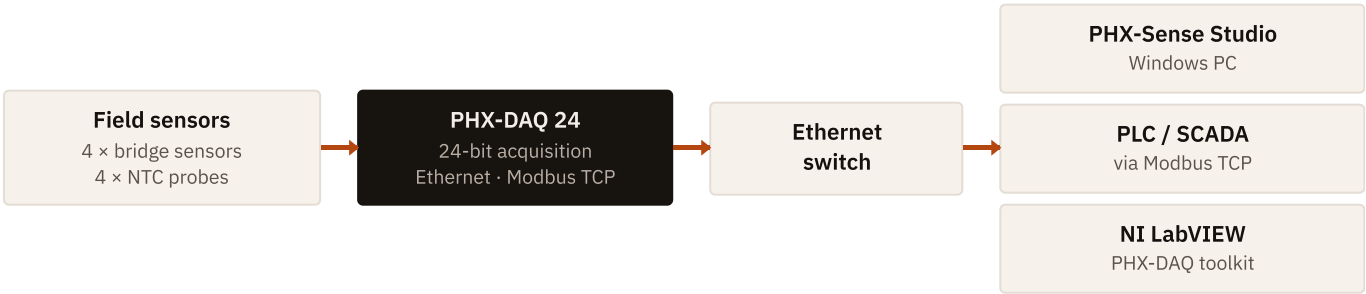
PHX-Sense Studio · PLC (Modbus TCP) · SCADA / HMI · NI LabVIEW · Custom software

TYPICAL APPLICATIONS

Bridges & flyovers · Dams & barrages · Buildings & towers · Tunnels & metro · Pile & load testing · Industrial weighing

Complete solution. Every unit is used with PHX-Sense Studio, our Windows console for discovery, live plots, calibration and CSV recording.

SYSTEM OVERVIEW



TECHNICAL SPECIFICATIONS

SENSOR INPUTS (BRIDGE-TYPE)	
Channels	4, differential, independent
Resolution	24-bit analog-to-digital converter, one per channel
Programmable gain	1, 2, 4, 8, 16, 32, 64, 128 (set per channel)
Full-scale input	±(1 ÷ gain) × excitation voltage. Example: ±7.8 mV/V at 128× gain
Sensor excitation	3.3 V DC per channel, ratiometric measurement: excitation drift cancels out. 350 Ω bridges recommended
Sensor configurations	Quarter, half and full bridge. Completion resistors are fitted externally, at the sensor
Sensor check	Detects an unconnected sensor or broken wire on each enabled channel. Runs at power-up, when a channel is switched on, and on request from PHX-Sense Studio (takes about 0.2 s)
Repeatability	±0.01% of reading, typical
Accuracy	±1% of reading, typical
TEMPERATURE INPUTS	
Channels / sensor	4 × NTC thermistor, 10 kΩ at 25 °C, β = 3950 (waterproof probes recommended)
Update rate	About 1 Hz per channel. A disconnected probe is reported as "no probe", not as a false reading
SAMPLING AND DATA	
Output rate	1 to 100 Hz, selectable (default 100 Hz). Each channel can be enabled or disabled
Data stream	Continuous TCP stream of every sample, to up to 3 receivers at the same time. Samples are sequence-numbered so a receiver can detect missing data
Live values (Modbus)	Per channel: averaged value in mV/V (IEEE-754 float) and raw code, plus temperatures and a status word (data fresh, overload, sensor check). Averaging window 1 to 100 samples, default 10, which removes 50 and 60 Hz mains pick-up at 100 Hz
COMMUNICATIONS	
Network	10/100 Mbit/s Ethernet, RJ45. Modbus TCP (port 502, up to 4 clients), TCP data stream (port 5000), automatic network discovery. Factory default: fixed IP 192.168.10.10, subnet mask 255.255.255.0. DHCP or another static IP can be set
Wi-Fi	Built-in 2.4 GHz access point (WPA2) for commissioning. Always on while the unit uses a fixed IP (the factory default); with DHCP it starts only when Ethernet is unavailable
Firmware update	Over the network from PHX-Sense Studio. The unit returns to the previous firmware automatically if an update does not complete correctly
Configuration security	Over the network, settings, restart, factory reset and firmware update require authentication. Measurement data is readable without authentication
Factory reset	From PHX-Sense Studio, or on site: with the unit running, hold the BOOT button 10 s until the LED turns solid white. Restores network settings, sample rate, channels, gains and access key. Calibration stored in PHX-Sense Studio is kept
POWER, ENVIRONMENTAL AND MECHANICAL	
Supply	24 V DC nominal (12 to 28 V DC), screw terminal. Fused, reverse-polarity and surge protected
Power consumption	Typically under 3 W at 24 V with all channels streaming
Operating temperature	0 to +50 °C
Enclosure	Aluminium enclosure recommended (see page 3). IP65 when installed in an IP65-rated enclosure with cable glands
Dimensions / weight	Depends on the enclosure supplied
Status LED	Blue: Ethernet connected. Green: access point active. Three quick blinks then a pause: ready. Single slow blink: a PC, PLC or LabVIEW is reading data. Slow red: no network. Fast white: reset button held; solid white: restoring factory defaults

Values are design values and may be refined after production testing.

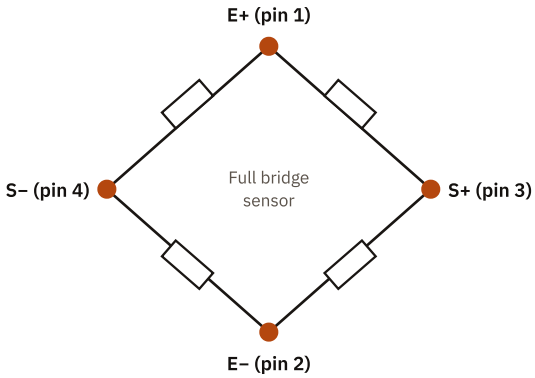
CONNECTIONS AND WIRING

SENSOR CONNECTOR (ONE PER CHANNEL)

6-position pluggable screw terminal, 5.08 mm pitch. Four identical connectors, CH1 to CH4.

Pin	Function
1	Excitation + (E+)
2	Excitation – (E–)
3	Signal + (S+)
4	Signal – (S–)
5	Temperature probe, lead 1
6	Temperature probe, lead 2 (no polarity)

BRIDGE SENSOR CONNECTION



OTHER CONNECTORS

Power	2-position pluggable screw terminal, 24 V DC
Network	RJ45, 10/100 Ethernet

WIRING NOTES

- **Full-bridge sensors** (load cells, pressure and force transducers) connect directly to pins 1 to 4.
- **Quarter- and half-bridge strain gauges** need bridge completion resistors. Fit them in a junction box close to the gauge and run the completed bridge to pins 1 to 4. Keeping completion at the sensor removes cable-resistance errors on long runs.
- Use **shielded, twisted-pair cable** and connect the shield at the logger end only.
- **Temperature probes** are two-wire, with no polarity. Waterproof probes are recommended for outdoor locations.
- Sensors must be suitable for a **3.3 V excitation**. Output in mV/V is unchanged, but the absolute signal is smaller than with a 5 V or 10 V supply, and the gain setting compensates for this.
- Do not connect sensor excitation lines to any external supply.
- Switch off unused channels in PHX-Sense Studio. An enabled channel with nothing connected is reported as "open".

RECOMMENDED ENCLOSURE

Use an aluminium enclosure (IP65 or better) with cable glands. It shields the millivolt-level sensor signals from electrical noise, spreads heat from the power stage and protects the unit on site.

- Mount the board on insulated standoffs. Do not let it touch the metal.
- Connect the enclosure to protective earth according to site practice.
- Bring every cable in through a gland. Ethernet, power and sensor cables included.
- A metal box blocks the built-in Wi-Fi. Commission over Ethernet, or fit an external antenna.

COMMISSIONING STEPS

1. Wire the sensors and probes to the sensor connectors with the power off.
2. Connect the Ethernet cable, then apply 24 V DC. The status LED turns blue when the network link is up.
3. A new unit uses the fixed address 192.168.10.10. Give the PC an address on the same subnet (for example 192.168.10.100), then start PHX-Sense Studio. The unit appears automatically in the Devices list.
4. Under Network setup, give the unit its own IP address (or DHCP). Every new unit starts at the same address, so set them up one at a time before connecting several to one network.
5. Check that no connected channel shows "OPEN", then set the unit name, sample rate, gains and channel calibration, and start recording.
6. Set a site-specific access key under Network setup before leaving the unit in service.

PHX-SENSE STUDIO

Windows console supplied with every PHX-DAQ 24 for setup, monitoring and recording. Requires Windows 10 or 11 (64-bit) and a network connection to the logger.

- Automatic discovery of every logger on the network
- Connect to one unit or many, each with its own settings
- Live multi-channel plots and numeric readouts
- Per-channel names, units, calibration and zero (tare)
- Selectable digital filters
- Shows "OPEN" on a channel with no sensor, with a one-click sensor check
- Recording to CSV, one file per module, with session notes and event markers
- Temperature logging alongside the sensor channels
- Sample rate, gain, channel selection and PLC averaging per unit
- Network setup, access key and factory reset
- Firmware update with progress and automatic rollback
- Unit health view: uptime and restart history

INTEGRATION WITH YOUR SYSTEMS

PHX-DAQ 24 exposes its measurements over standard Modbus TCP, so PLCs, SCADA systems, HMIs and custom software can read them without extra hardware or drivers.

PLC · SCADA · HMI

Read the live values with a standard Modbus TCP client: averaged mV/V per channel as 32-bit floats, temperatures in °C and a status word. Scale to load, strain or pressure in the PLC.

NI LABVIEW

The PHX-DAQ LabVIEW toolkit provides ready-made VIs to find modules, monitor live values and log every sample to CSV, plus a .NET driver for your own VIs.

CUSTOM SOFTWARE

Any language with Modbus TCP support can poll the live values. A continuous TCP data stream delivers every sample for high-rate acquisition.

- Tested with generic Modbus TCP clients, QModMaster and NI LabVIEW.
- The Modbus register map, stream format and LabVIEW toolkit are provided to customers and integration partners on request.
- Measurement and status data can be read by any Modbus TCP client. Changing settings requires authentication.
- Up to 3 programs can receive the full-rate data stream at once, and up to 4 can poll Modbus at the same time.

NETWORK SECURITY GUIDANCE

- Install the logger on a **trusted, private network or VLAN**. Do not expose it directly to the internet.
- Measurement data is readable by any device on the same network segment. Use network segmentation where data confidentiality matters.
- Set a **site-specific access key** at commissioning and store it securely. A lost key is cleared with the reset button, which anyone with physical access can use: install the unit in a **locked enclosure**.
- Change the Wi-Fi access-point password if the access point is left enabled.
- Keep firmware up to date. Phoenixtek issues updates through PHX-Sense Studio.

ORDERING INFORMATION

Part	PHX-DAQ 24 — 4-channel sensor and temperature datalogger, with PHX-Sense Studio
Accessories	NTC temperature probes, bridge-completion junction boxes, shielded cable, 24 V DC power supply. Available on request
Software & support	PHX-Sense Studio, LabVIEW toolkit, Modbus integration guide, calibration guidance and pilot deployments

PHOENIXTEK SOLUTIONS

Request a demo or a pilot deployment

www.phoenixtek.in
info@phoenixtek.in

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